Infinite Series 5

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Problem 1

Take any $\alpha = a + b\sqrt{-d}$ with $a, b \in \mathbb{Q}$. For α to be integral, $(x-\alpha)(x-\bar{\alpha})$ must have integer coefficients. We have that

$$\begin{aligned}(x-\alpha)(x+\alpha) &= (x-(a+b\sqrt{-d}))(x-(a-b\sqrt{-d})) \\ &= x^2 - 2ax + a^2 + db^2,\end{aligned}$$

so $2a$ and $a^2 + db^2$ must be integers. Let $u = 2a$ and $v = 2b$. We have that u must be an integer. Now, consider $4(a^2 + db^2)$:

$$4(a^2 + db^2) = (2a)^2 + d(2b)^2 = u^2 + dv^2.$$

That must be an integer, and since u is an integer, $v = 2b$ must be an integer as well. So every integral element has the form

$$\alpha = \frac{u + v\sqrt{-d}}{2}, \quad u, v \in \mathbb{Z}.$$

Now, we need to find which u, v give integers. For $a^2 + db^2$ to be an integer,

$$\frac{u^2 + dv^2}{4} \in \mathbb{Z} \iff u^2 \equiv -dv^2 \pmod 4.$$

If v is even, then $v^2 \equiv 0 \pmod 4$, so u must be even. If v is odd, then $v^2 \equiv 1 \pmod 4$, so u^2 is not even and $u^2 \equiv 1 \pmod 4$. So we need $-d \equiv 1 \pmod 4$. If $-d \not\equiv 1 \pmod 4$, no integral element can have v odd, so α must lie in $\mathbb{Z}[\sqrt{-d}]$.

Problem 5

Define multiplication on classes by $[a] \cdot [b] = [ab]$.
We need to show that if $a \sim a'$ and $b \sim b'$ then $ab \sim a'b'$. Assume that $(c)a = (d)a'$ and $(e)b = (f)b'$ for some $c, d, e, f \in R$. Then

$$(ce)(ab) = ((c)a)((e)b) = ((d)a')((f)b') = (df)(a'b').$$

So $(ce)(ab) = (df)(a'b')$, so $ab \sim a'b'$. Next, we need to show associativity. We know that $(ab)c = a(bc)$, so

$$([a] \cdot [b]) \cdot [c] = [ab] \cdot [c] = [(ab)c] = [a(bc)] = [a] \cdot [bc] = [a] \cdot ([b] \cdot [c]).$$

Problem 8

Each term in the product is the sum of a geometric series:

$$\prod_{p \text{ prime}} \frac{1}{1 - \frac{\chi(p)}{p^s}} = \prod_{p \text{ prime}} \sum_{n=0}^{\infty} \left(\frac{\chi(p)}{p^s} \right)^n.$$

Expanding the product term-by-term, it sums over each unique prime factorization, which is just all natural numbers, and χ is multiplicative, so the product is

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$